Yuchen Miao

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Education

Northeastern University, China 2023 - 2027 (expected)

Sydney Smart Technology College | B.Eng. in Communication Engineering (joint program)

GPA: 3.8925/5.0 | Average: 88.95/100 | Rank: 16/108 | IELTS: 7.0

Research Interests

Agents; LLM applications; recommender systems; multimodal fusion.

Publications

Peer-reviewed / accepted

[1] **Yuchen Miao**, Zijun Wang, Chang Han, Yurui Shi, Mingtai Zhang, and Siyang Xu. "The '10th Juror': Open-Set Standpoint Screening for Bureaucratic Bias Detection". **EMNLP 2026, Main Conference** [ICORE A*; CCF-B]. [Paper].

[2] **Yuchen Miao**, Zijun Wang, Ke Liu, and Siyang Xu. "Anatomy of a Decision: Uncertainty-aware Hierarchical Intent Learning via Flow Matching for Multimodal Recommendation". **WISE 2026** [ICORE B; CCF-B]. [Paper].

[3] **Yuchen Miao**, Zijun Wang, Ke Liu, Peixuan Wang, and Chang Han. "IMEX-FND: A Traceable Interaction-Aware Mixture-of-Experts Framework for Multimodal Fake News Detection". **WISE 2026** [ICORE B; CCF-B]. [Paper].

[4] **Yuchen Miao**, Mingxuan Cui, Yitong Zhu, Yu Wang, and Siyang Xu. "R3-REC: Reasoning-Driven Recommendation via Retrieval-Augmented LLMs over Multi-Granular Interest Signals". **ICASSP 2026** [CCF-B]. [Paper].

[5] Yurui Shi*, **Yuchen Miao***, Ximing Hu, Zijun Wang, and Chang Han. "MOTIF: Motivation-guided Topology Inference for Cold-start Multimodal Recommendation". **WISE 2026** [ICORE B; CCF-B]. [Paper] *Equal contribution.

Preprint

[6] **Yuchen Miao**. "Universal Refinement without Interaction: Order-Optimal 1-Bit Mean Estimation". arXiv:2607.24358, 2026. [Paper].

Research Experience

Undergraduate Researcher 2025 - present

Hebei Key Laboratory of Marine Perception Network and Data Processing

Developed evidence-grounded agent and multimodal systems, with work spanning method design, implementation, controlled experiments, component ablations, and paper writing.

MARS-Gov: Evidence-Grounded Multi-Agent Review EMNLP 2026

- Orchestrated a Scout, nine fixed jurors, and temporary jurors for uncovered groups. Defined JSON output contracts for review labels, confidence, and supporting cues; passed raw reports and shared evidence to a Judge for adjudication.
- Built a three-stage retrieval pipeline combining keyword search, dense retrieval, and adjacent-chunk expansion. Supplied neutral evidence summaries to downstream agents and retained source identifiers in decision traces.
- Implemented rule-based routing that clears low-risk cases only when outputs are well formed, and escalates risk alerts or missing/malformed reports to the Judge, limiting unnecessary adjudication and rewriting calls.
- Integrated minimal-edit rewriting with a detection-only verification pass using the same bias criteria and disabling further rewriting. Recorded evidence, agent reports, routing actions, and verification outcomes for audit.
- Evaluated detection, held-out-group recovery, and intervention quality separately, with component ablations and token-cost analysis. Achieved F1 = 0.880 on DGDB and Correct@1 = 85.1% in leave-one-category-out evaluation.

Non-Adaptive One-Bit Mean Estimation Preprint

- Developed a fully non-adaptive public-coin protocol for scalar mean estimation under known mean-range and central-moment bounds. Used decoder-side refinement to match adaptive sample-complexity rates in the small-error, high-confidence regime.

Selected Research Contributions

MMFact: Multimodal Misinformation Detection Toolbox [Code] [arXiv]

- Co-developed a unified toolbox integrating 47 detector implementations and baselines with shared datasets and reusable preprocessing resources.
- Contributed to configurable training and evaluation workflows with validation-based model selection, multi-seed aggregation, and class- and domain-level reporting, while preserving method-specific inputs and training objectives.

Retrieval and LLM Workflow Design

- **R3-REC (ICASSP 2026):** Built retrieval-augmented LLM ranking from multi-granular intents, preference polarity, item keyphrases, and BM25/Sentence-BERT similar-user retrieval. Evaluated ranking quality, component ablations, and inference latency.
- **MOTIF (WISE 2026):** Co-developed an offline LLM pipeline that infers user motivations and reconstructs item topology for cold-start graph learning, retaining graph-only online inference. Achieved up to 6.07% relative improvement in Recall@20/NDCG@20 over the best compared baseline.

Uncertainty and Multimodal Representations WISE 2026

- **UHIFlow:** Used conditional flow matching to construct modality-specific uncertainty scores and adapt the depth of user-intent hierarchies. Achieved 2.81%-3.04% relative gains in Recall/NDCG over the best compared baseline for each metric across three datasets.
- **IMEX-FND:** Designed sample-level routing and modality-replacement constraints to distinguish modality-specific, synergistic, and redundant information. Across three datasets, removing routing reduced Macro-F1 by 3.1 percentage points on average on subsets where text-only and image-only predictions disagreed.

Research Program & Patent

National College Students' Innovation and Entrepreneurship Training Program

Project No. 202619145026 2025 - present

Research on interpretable recommendation and fake-content detection for new media security; contributed to R3-REC and IMEX-FND.

Yuchen Miao, Yu Wang, Gang Wang, Xin Feng, and Siyang Xu. "Explainable Multi-Domain Mixture-of-Experts Method for Multimodal Fake-Content Detection." Chinese invention patent application No. 202511816918.8; filed December 4, 2025; in substantive examination.

Industry Experience

China Development Bank, Shanxi Branch | IT Intern Jul - Aug 2025

Huajin Coking Coal Co., Ltd. | Big Data Center Intern Jul - Sep 2024

Beijing Codnoy Technology Co., Ltd. | Engineering Intern Jul - Oct 2024

Selected Honors & Technical Skills

Honors: Finalist, Mathematical Contest in Modeling (MCM), 2026; Northeastern University Third-Class Scholarship (five awards, 2023 - 2026); National Third Prize, Advanced Robot and Simulation Technology Competition, 2025; Provincial Second Prize, National College Students' Market Research & Analysis Competition, 2025.

Programming & tools: Python, PyTorch, C, MATLAB, LaTeX.

Agent systems: Role orchestration, structured inter-agent outputs, retrieval and evidence grounding, risk-based routing, and rewrite verification.

Modeling & evaluation: Multimodal learning, flow matching, mixture-of-experts, graph learning, component ablations, and quality/cost evaluation.